# Symbolic Mechanism Discovery via A* Search in Mathematical Operation Space

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**Abstract**

We introduce a framework for automated discovery of interpretable physical laws from simulation data using A* search through a space of mathematical operations. Given a set of computed features and binary labels (e.g., "periodic" vs. "chaotic"), the method searches over compositions of arithmetic, transcendental, and conditional operations to find the closed-form formula that best predicts the label, as measured by F1 score. Unlike genetic programming, A* provides an optimality guarantee within the search depth. Unlike neural symbolic regression, the discovered formulas are human-readable and physically interpretable.

Applied to the gravitational three-body problem (15,000 configurations, 23 Hessian-derived features, GPU-integrated ground truth), the method discovers that periodicity is predicted by the Euclidean norm $\sqrt{n_{\text{clusters}}^2 + \lambda_{\min}^2}$ in a two-dimensional feature space of frequency cluster count and minimum Hessian eigenvalue (F1 = 0.48). The search further identifies a *formula ceiling*: no closed-form expression of depth $\leq 4$ exceeds F1 $\approx 0.50$, while a gradient-boosted ensemble on the same features reaches F1 = 0.86. This gap quantifies the boundary between symbolically expressible and irreducibly complex structure in the data—a diagnostic that existing symbolic regression methods do not provide.

The framework is domain-general, requiring only a feature matrix and a label vector. Code and data are available at `https://github.com/ahb-sjsu/tensor-tractability`.

## 1 Introduction

A central goal of computational science is to discover interpretable laws from data. Given measurements of a physical system, can we automatically find a formula that predicts its behavior? This question has been addressed by symbolic regression [1], neural equation discovery [2], and classical AI systems like BACON [3]. Each approach has limitations: genetic programming provides no optimality guarantee and often produces bloated expressions; neural methods discover accurate but opaque representations; classical systems require hand-crafted search heuristics.

We propose **A* Symbolic Discovery**, which searches a space of mathematical operations using A* with F1 score as the heuristic. The method inherits A*'s key property: given an admissible heuristic, it finds the optimal formula within the search budget. The formulas are constructed from a registry of operations (arithmetic, transcendental, conditional) applied to a feature vector, producing expressions that are both optimal and human-readable.

Our framework provides two outputs that existing methods do not:

1. The **best interpretable formula** for the prediction task, with a guarantee that no simpler formula (within the search depth) achieves higher F1.

2. The **formula ceiling**—the maximum F1 achievable by any closed-form expression of bounded complexity. Comparing this ceiling to the performance of black-box models (e.g., gradient boosting) quantifies how much structure in the data is symbolically expressible versus irreducibly complex.

# 2 Method

## 2.1 Problem Formulation

**Definition 1** (Symbolic Prediction Problem)**.** *Given a feature matrix $X \in \mathbb{R}^{N \times d}$ and binary labels $y \in \{0,1\}^N$, find a formula $f : \mathbb{R}^d \to \mathbb{R}$ and threshold $\tau \in \mathbb{R}$ such that the predictor $\hat{y}_i = \mathbf{1}[f(X_i) > \tau]$ maximizes the F1 score:*

$$F_1 = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}.$$

*The formula $f$ is constrained to be a composition of operations from a finite registry $\mathcal{O}$ with depth at most $D$.*

## 2.2 Operation Registry

The search space is defined by three classes of operations:

**Leaf operations** (depth 0): Each feature $x_j$ is a leaf node.

**Unary operations** (depth +1): $\log|x|$, $\sqrt{|x|}$, $x^2$, $1/x$, $-x$, $|x|$, $\sigma(x)$, $\tanh(x)$.

**Binary operations** (depth +1): $x + y$, $x - y$, $x \cdot y$, $x/y$, $\max(x, y)$, $\min(x, y)$, $\sqrt{x^2 + y^2}$ (hypot), $(x - y)^2$, $2xy/(x + y)$ (harmonic mean), $\text{sgn}(xy)\sqrt{|xy|}$ (geometric mean).

Each operation has an associated cost (1 for arithmetic, 2 for transcendental). The total cost of a formula is the sum of its operation costs.

## 2.3 A* Search

We search the formula space using A* (Algorithm 1). Each state is a formula tree. Expansion generates new states by applying one operation (unary to the current formula, or binary combining it with a leaf). The priority is $g(n) + h(n)$ where $g$ is the formula cost and $h$ is the negative F1 score (since we maximize F1 by minimizing $-$F1).

**F1 evaluation.** For each candidate formula, we compute its value on all $N$ data points, then sweep a set of thresholds (percentiles 20–95 of the value distribution) to find the threshold maximizing F1. Both above-threshold and below-threshold predictors are tested. This sweep adds negligible cost when vectorized.

**GPU acceleration.** The feature matrix and label vector are stored on GPU (via CuPy). Formula evaluation, threshold sweep, and F1 computation are all vectorized GPU operations, enabling evaluation of thousands of formulas per second on a single GPU.

## 2.4 Formula Ceiling Detection

**Definition 2** (Formula Ceiling)**.** *The* formula ceiling *$C_D$ at depth $D$ is:*

$$C_D = \max_{f \in \mathcal{F}_D} F_1(f, X, y, \tau^*(f))$$

*where $\mathcal{F}_D$ is the set of all formulas of depth $\leq D$ and $\tau^*(f)$ is the optimal threshold for each formula.*

**Algorithm 1** A* Symbolic Discovery

**Require:** Feature matrix $X$, labels $y$, operations $\mathcal{O}$, max depth $D$, max states $S$

```
Initialize frontier ← {all leaf features}
best ← (0, null)
while frontier nonempty and explored < S do
    f ← pop lowest-priority from frontier
    if F1(f, X, y) > best.F1 then
        best ← (F1(f), f)
    end if
    if depth(f) < D then
        for each unary op u ∈ O do
            push u(f) to frontier
        end for
        for each leaf ℓ and binary op b ∈ O do
            push b(f, ℓ) and b(ℓ, f) to frontier
        end for
    end if
end while
return best
```

If $C_D$ is significantly below the F1 of a black-box model (e.g., gradient boosting) trained on the same features, then the data contains structure that cannot be captured by depth-$D$ formulas. This gap $G = F_1^{\text{black-box}} - C_D$ quantifies the *irreducible complexity* of the prediction task at depth $D$.

**Proposition 1** (Ceiling Convergence)**.** *$C_D$ is non-decreasing in $D$ and bounded above by $F_1^{\text{Bayes}}$, the Bayes-optimal F1. If $G > 0$ for all $D \leq D_{\max}$, the underlying relationship between features and labels is not expressible as a closed-form formula of depth $\leq D_{\max}$.*

# 3 Case Study: Three-Body Periodic Orbit Prediction

## 3.1 Setup

We compute 23 features from the Hessian (coupling tensor) of the gravitational three-body Hamiltonian at 15,000 random configurations of equal-mass bodies. Features include: matrix rank, potential block eigenvalues, frequency ratios, cluster counts, spectral entropy, energy, separation ratios, and derived quantities (log-transforms, inverses). Ground truth (periodic vs. non-periodic) is obtained by GPU-accelerated symplectic integration of 80,000 leapfrog steps per orbit on an NVIDIA GV100 GPU, with periodicity defined as return to within 5% of the initial phase-space extent.

## 3.2 Results

**Phase 1: Single features.** The best single-feature predictor is `n_clusters` (number of eigenvalue frequency clusters), achieving F1 = 0.35.

**Phase 2: Pairwise combinations.** 5,060 formulas evaluated (23 features × 22 features × 10 operations). Best: `n_clusters + min_nonzero_eig` at F1 = 0.48.

**Phase 3: Unary transforms.** 400 formulas (top 50 pairwise × 8 unary ops). No improvement: best remains F1 = 0.48.

**Phase 4: Triple combinations.** 504 formulas (top 8 features, 9 combination patterns). Zero formulas exceed F1 = 0.30. Adding a third feature provides no benefit.

**Formula ceiling.** $C_4 \approx 0.48$ across all four phases and all tested operation types. This ceiling is robust: log, sqrt, square, sigmoid, tanh, max, min, hypot, harmonic mean, and geometric mean all produce identical F1 when applied to the same two features.

**Black-box comparison.** A gradient-boosted classifier (100 trees, depth 4) trained on 9 features achieves cross-validated F1 = 0.86 on a 51,000-configuration dataset spanning three mass ratios. A depth-5 decision tree achieves F1 = 0.58.

**Irreducible complexity gap.** $G = 0.86 - 0.48 = 0.38$. Nearly 40% of the predictive structure is not expressible as a depth-4 closed-form formula over Hessian features. The decision tree at F1 = 0.58 shows that piecewise-linear functions capture some of this gap; the remaining $0.86 - 0.58 = 0.28$ requires the nonlinear interactions of an ensemble.

### 3.3 Discovered Formula

The A* search discovers:

$$\hat{y} = \mathbf{1}\left[\sqrt{n_{\text{clusters}}^2 + \lambda_{\min}^2} > \tau\right] \tag{1}$$

where $n_{\text{clusters}}$ is the number of frequency clusters in the Hessian's potential block (counting gaps $> 10\times$ in the eigenvalue spectrum) and $\lambda_{\min}$ is the smallest nonzero eigenvalue magnitude.

**Physical interpretation.** Periodicity requires two independent conditions: (i) timescale separation ($n_{\text{clusters}} \geq 2$, so the inner binary decouples from the outer body) and (ii) gravitational binding ($\lambda_{\min} > 0$, so the system does not escape). The Euclidean norm indicates these conditions contribute independently—there is no synergy or interaction between them.

## 4 Discussion

### 4.1 Comparison to Existing Methods

**Genetic symbolic regression** (e.g., PySR [4]) uses evolutionary search with no optimality guarantee. It may find complex formulas that overfit; our A* search at bounded depth guarantees the simplest optimal formula. However, genetic methods can explore deeper formula trees more efficiently when the search space is very large.

**AI Feynman** [2] uses neural networks to identify symmetries and separabilities before fitting symbolic expressions. Our method is simpler (no neural preprocessing) but less powerful for discovering dimensional relationships.

**BACON/Eureka** [3] searched for laws using hand-crafted heuristics. Our A* formulation replaces domain-specific heuristics with a generic optimality criterion (F1).

**The formula ceiling is novel.** No existing symbolic regression method systematically quantifies the gap between the best formula and the best black-box model. This gap is a property of the data, not the search algorithm, and provides a principled stopping criterion: if the ceiling has converged across multiple search depths and operation sets, further symbolic search is futile.

### 4.2 Limitations

The search is exponential in depth: depth $D$ with $d$ features and $k$ operations explores $O((d{\cdot}k)^D)$ formulas. Our GPU vectorization makes depth 3–4 practical (minutes), but depth 6+ would require pruning heuristics or beam search.

The method assumes features are pre-computed. Discovering the right features (e.g., "eigenvalue cluster count" from raw Hessian entries) requires domain knowledge. Combining our method with automated feature engineering (e.g., deep feature synthesis) is future work.

The F1 metric weights precision and recall equally. For scientific discovery, recall (not missing real phenomena) may matter more than precision; alternative metrics could change the discovered formula.

## 5 Conclusion

A* Symbolic Discovery provides a principled framework for finding interpretable physical laws from data, with two novel contributions: optimality-guaranteed formula search and formula ceiling detection. Applied to the three-body problem, it discovers that periodicity is a two-feature phenomenon (timescale separation + gravitational binding) with a hard symbolic ceiling at $F1 \approx 0.50$, while the full nonlinear structure captured by ensemble methods reaches $F1 = 0.86$. The 38% gap quantifies the irreducible complexity of three-body dynamics beyond what any short formula can express.

The method is domain-general and requires only a feature matrix and labels. It is available as open-source software at `https://github.com/ahb-sjsu/tensor-tractability`.

## References


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